Insurance Portfolio Risk & Profitability Analysis

The objective of this analysis was to evaluate the profitability and risk exposure of the insurance portfolio by comparing premium income against claims payouts across farms and branches. The goal was to identify loss-making farms, revenue concentration risk, and areas requiring premium reassessment or operational review to support management decision-making. *This project involved analyzing a large insurance dataset containing 104 columns and approximately 22,090 rows from multiple company branches. The tasks included data cleaning, sorting, premium calculation for each animal (3% of Sum Insured), identifying farms with the highest tag numbers and initial sum insured, and comparing yearly claims data (2024 vs 2025) to determine business volume. The analysis also included creation of visualizations: a pie chart showing percentage contributions of the top 5 farms to total premium income and a bar chart comparing total claims paid versus total premiums collected. Additionally, the claims-to-premium ratio was calculated to identify the most efficient farm. The workflow demonstrates handling of large datasets, analytical calculations, and visualization in Excel/Google Sheets to extract actionable insights.*

Data Cleaning Process

Data Cleaning & Preparation

- Removed irrelevant/empty columns and rows to ensure dataset integrity.
- Standardized formats and handled missing values to enable reliable calculations.
- Tag & Premium Management
- Consolidated Tag Number/Date, extracted numeric values, and computed premiums.
- Calculated totals, ratios, and highest values for actionable insights.

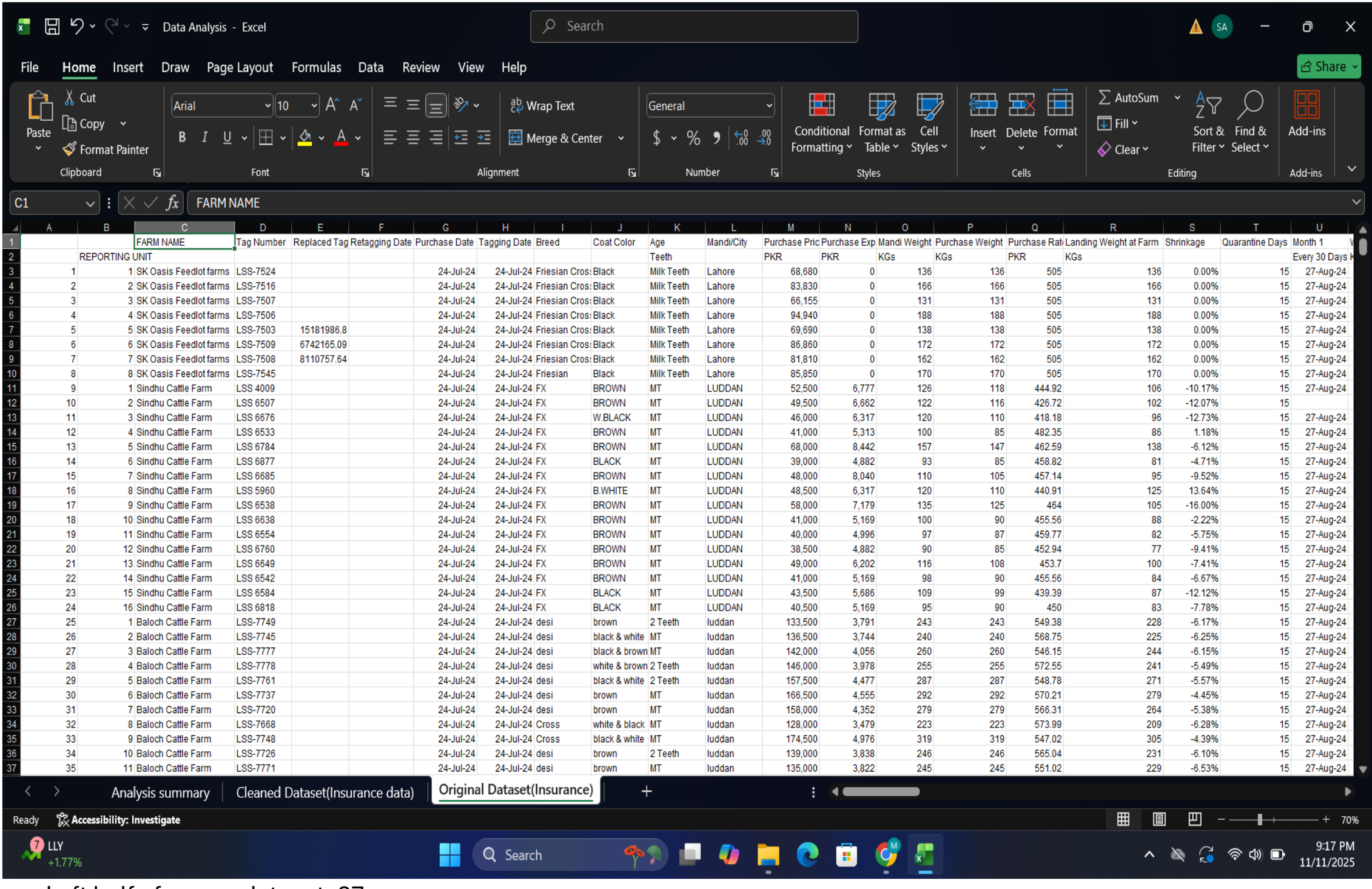
Analysis & Visualization

- Generated farm-level summaries and year-over-year business trends.
- Created charts to highlight top contributors and claims performance.

Technical Skills

- Advanced Excel: SUMIF, IFERROR, REGEXEXTRACT, pivot tables
- Data cleaning, filtering, formatting, and aggregation
- Financial calculations and data visualization

Messy Dataset (Before Cleaning)



		FARM NAME	Tag Number	Replaced Tag	Retagging Date	Purchase Date	Tagging Date	Breed	Coat Color	Age	Mandi/City	Purchase Pric	Purchase Exp	Mandi Weight	Purchase Weight	Purchase Rat	Landing Weight at Farm	Shrinkage	Quarantine Days	Month 1
	REPORTING UNIT									Teeth		PKR	PKR	KGs	KGs	PKR	KGs			Every 30 Days
1	1	SK Oasis Feedlot farms	LSS-7524			24-Jul-24	24-Jul-24	Friesian Cros	Black	Milk Teeth	Lahore	68,680	0	136	136	505	136	0.00%	15	27-Aug-24
2	2	SK Oasis Feedlot farms	LSS-7516			24-Jul-24	24-Jul-24	Friesian Cros	Black	Milk Teeth	Lahore	83,830	0	166	166	505	166	0.00%	15	27-Aug-24
3	3	SK Oasis Feedlot farms	LSS-7507			24-Jul-24	24-Jul-24	Friesian Cros	Black	Milk Teeth	Lahore	66,155	0	131	131	505	131	0.00%	15	27-Aug-24
4	4	SK Oasis Feedlot farms	LSS-7506			24-Jul-24	24-Jul-24	Friesian Cros	Black	Milk Teeth	Lahore	94,940	0	188	188	505	188	0.00%	15	27-Aug-24
5	5	SK Oasis Feedlot farms	LSS-7503	15181986.8		24-Jul-24	24-Jul-24	Friesian Cros	Black	Milk Teeth	Lahore	69,690	0	138	138	505	138	0.00%	15	27-Aug-24
6	6	SK Oasis Feedlot farms	LSS-7509	6742165.09		24-Jul-24	24-Jul-24	Friesian Cros	Black	Milk Teeth	Lahore	86,860	0	172	172	505	172	0.00%	15	27-Aug-24
7	7	SK Oasis Feedlot farms	LSS-7508	8110757.64		24-Jul-24	24-Jul-24	Friesian Cros	Black	Milk Teeth	Lahore	81,810	0	162	162	505	162	0.00%	15	27-Aug-24
8	8	SK Oasis Feedlot farms	LSS-7545			24-Jul-24	24-Jul-24	Friesian	Black	Milk Teeth	Lahore	85,850	0	170	170	505	170	0.00%	15	27-Aug-24
9	1	Sindhu Cattle Farm	LSS 4009			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	52,500	6,777	126	118	444.92	106	-10.17%	15	27-Aug-24
10	2	Sindhu Cattle Farm	LSS 6507			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	49,500	6,662	122	116	426.72	102	-12.07%	15	
11	3	Sindhu Cattle Farm	LSS 6676			24-Jul-24	24-Jul-24	FX	W.BLACK	MT	LUDDAN	46,000	6,317	120	110	418.18	96	-12.73%	15	27-Aug-24
12	4	Sindhu Cattle Farm	LSS 6533			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	41,000	5,313	100	85	482.35	86	1.18%	15	27-Aug-24
13	5	Sindhu Cattle Farm	LSS 6784			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	68,000	8,442	157	147	462.59	138	-6.12%	15	27-Aug-24
14	6	Sindhu Cattle Farm	LSS 6877			24-Jul-24	24-Jul-24	FX	BLACK	MT	LUDDAN	39,000	4,882	93	85	458.82	81	-4.71%	15	27-Aug-24
15	7	Sindhu Cattle Farm	LSS 6685			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	48,000	8,040	110	105	457.14	95	-9.52%	15	27-Aug-24
16	8	Sindhu Cattle Farm	LSS 5960			24-Jul-24	24-Jul-24	FX	B.WHITE	MT	LUDDAN	48,500	6,317	120	110	440.91	125	13.64%	15	27-Aug-24
17	9	Sindhu Cattle Farm	LSS 6538			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	58,000	7,179	135	125	464	105	-16.00%	15	27-Aug-24
18	10	Sindhu Cattle Farm	LSS 6638			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	41,000	5,169	100	90	455.56	88	-2.22%	15	27-Aug-24
19	11	Sindhu Cattle Farm	LSS 6554			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	40,000	4,996	97	87	459.77	82	-5.75%	15	27-Aug-24
20	12	Sindhu Cattle Farm	LSS 6760			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	38,500	4,882	90	85	452.94	77	-9.41%	15	27-Aug-24
21	13	Sindhu Cattle Farm	LSS 6649			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	49,000	6,202	116	108	453.7	100	-7.41%	15	27-Aug-24
22	14	Sindhu Cattle Farm	LSS 6542			24-Jul-24	24-Jul-24	FX	BROWN	MT	LUDDAN	41,000	5,169	98	90	455.56	84	-6.67%	15	27-Aug-24
23	15	Sindhu Cattle Farm	LSS 6584			24-Jul-24	24-Jul-24	FX	BLACK	MT	LUDDAN	43,500	5,686	109	99	439.39	87	-12.12%	15	27-Aug-24
24	16	Sindhu Cattle Farm	LSS 6818			24-Jul-24	24-Jul-24	FX	BLACK	MT	LUDDAN	40,500	5,169	95	90	450	83	-7.78%	15	27-Aug-24
25	1	Baloch Cattle Farm	LSS-7749			24-Jul-24	24-Jul-24	desi	brown	2 Teeth	luddan	133,500	3,791	243	243	549.38	228	-6.17%	15	27-Aug-24
26	2	Baloch Cattle Farm	LSS-7745			24-Jul-24	24-Jul-24	desi	black & white	MT	luddan	136,500	3,744	240	240	568.75	225	-6.25%	15	27-Aug-24
27	3	Baloch Cattle Farm	LSS-7777			24-Jul-24	24-Jul-24	desi	black & brown	MT	luddan	142,000	4,056	260	260	546.15	244	-6.15%	15	27-Aug-24
28	4	Baloch Cattle Farm	LSS-7778			24-Jul-24	24-Jul-24	desi	white & brown	2 Teeth	luddan	146,000	3,978	255	255	572.55	241	-5.49%	15	27-Aug-24
29	5	Baloch Cattle Farm	LSS-7761			24-Jul-24	24-Jul-24	desi	black & white	2 Teeth	luddan	157,500	4,477	287	287	548.78	271	-5.57%	15	27-Aug-24
30	6	Baloch Cattle Farm	LSS-7737			24-Jul-24	24-Jul-24	desi	brown	MT	luddan	166,500	4,555	292	292	570.21	279	-4.45%	15	27-Aug-24
31	7	Baloch Cattle Farm	LSS-7720			24-Jul-24	24-Jul-24	desi	brown	MT	luddan	158,000	4,352	279	279	566.31	264	-5.38%	15	27-Aug-24
32	8	Baloch Cattle Farm	LSS-7668			24-Jul-24	24-Jul-24	Cross	white & black	MT	luddan	128,000	3,479	223	223	573.99	209	-6.28%	15	27-Aug-24
33	9	Baloch Cattle Farm	LSS-7748			24-Jul-24	24-Jul-24	Cross	black & white	MT	luddan	174,500	4,976	319	319	547.02	305	-4.39%	15	27-Aug-24
34	10	Baloch Cattle Farm	LSS-7726			24-Jul-24	24-Jul-24	desi	brown	2 Teeth	luddan	139,000	3,838	246	246	565.04	231	-6.10%	15	27-Aug-24
35	11	Baloch Cattle Farm	LSS-7771			24-Jul-24	24-Jul-24	desi	brown	MT	luddan	135,000	3,822	245	245	551.02	229	-6.53%	15	27-Aug-24

Left half of messy dataset, 37 rows

Data Analysis - Excel

C1 | FARM NAME

	CH	CI	CJ	CK	CL	CM	CN	CO	CP	CQ	CR	CS	CT	CU	CV	CW	CX	CY
1	Reported to th	Weight at Farr	Farm Exit Date	Live Cattle Sale Amount	Buyer Name	Slaughter Location	Rejection	Sale Date	Total Carcass	Rejected Weig	Rejected Weight Rate per KG	Accepted Weight Rate per KG	Net Offal Sale Amount	Sale Expenses	Carcass Sale	Net Payment	Payment Date	Animal days
2	Date	KGs		PKR			Yes/No		KGs	KGs	PKR	PKR	PKR	PKR	PKR	PKR		
3		231	12-Oct-24		SK LiveStock	Raiwind	Yes	13-Oct-24	111	54	700	1,115	5,000	2,162	101,355	104,193		81
4		252	27-Oct-24	0	SK LiveStock	Raiwind	No	28-Oct-24	120	0	0	1,100	5,000	2,506	132,000	134,494		96
5		215	12-Oct-24		SK LiveStock	Raiwind	Yes	13-Oct-24	103	25	600	1,115	5,000	846	101,970	106,124		81
6		232	12-Oct-24		SK LiveStock	Raiwind	No	13-Oct-24	113	0	0	1,115	5,000	2,171	125,995	128,824		81
7		225	1-Nov-24	0	SK LiveStock	Raiwind	No	2-Nov-24	116	0	0	1,100	5,000	803	127,600	131,797		101
8		264	29-Oct-24		SK LiveStock	Raiwind	No	29-Oct-24	128	0	0	1,100	6,200	907	140,800	146,093		98
9		247	29-Oct-24		SK LiveStock	Raiwind	No	29-Oct-24	116	0	0	1,100	6,200	848	127,600	132,952		98
10		256	12-Oct-24		SK LiveStock	Raiwind	No	13-Oct-24	118	0	0	1,115	5,000	2,396	131,570	134,174		81
11		162	26-Oct-24	0	SK Livestock	Raiwind	No	27-Oct-24	71	0	0	1,100	4,500	1,611	78,100	80,989		95
12							No			0	0				0	0		16
13		181	26-Oct-24	0	SK Livestock	Raiwind	No	27-Oct-24	82	0	0	1,100	4,500	1,800	90,200	92,900		95
14		163	13-Oct-24	0	SK Livestock	Raiwind	No	13-Oct-24	76	0	0	1,115	4,500	1,525	84,740	87,715		82
15		205	26-Oct-24	0	SK Livestock	Raiwind	No	27-Oct-24	92	0	0	1,100	4,500	2,038	101,200	103,662		95
16		145	26-Oct-24	0	SK Livestock	Raiwind	No	27-Oct-24	67	0	0	1,100	4,500	1,442	73,700	76,758		95
17		168	26-Oct-24	0	SK Livestock	Raiwind	No	27-Oct-24	84	0	0	1,100	4,500	1,671	92,400	95,229		95
18		185	13-Oct-24	0	SK Livestock	Raiwind	No	13-Oct-24	84	0	0	1,115	4,500	1,731	93,660	96,429		82
19		185	13-Oct-24	0	SK Livestock	Raiwind	No	13-Oct-24	88	0	0	1,115	4,500	1,731	98,120	100,889		82
20		193	18-Nov-24	0	SK LiveStock	Raiwind	No	21-Nov-24	91.79	0	0	1,100	5,000	1,045	100,969	104,924		118
21		154	26-Oct-24	0	SK Livestock	Raiwind	No	27-Oct-24	72	0	0	1,100	4,500	1,531	79,200	82,169		95
22		156	13-Oct-24	0	SK Livestock	Raiwind	No	13-Oct-24	69	0	0	1,115	4,500	1,460	76,935	79,975		82
23		180	13-Oct-24	0	SK Livestock	Raiwind	No	13-Oct-24	85	0	0	1,115	4,500	1,685	94,775	97,590		82
24		155	13-Oct-24	0	SK Livestock	Raiwind	No	13-Oct-24	72	0	0	1,115	4,500	1,451	80,280	83,329		82
25		164	13-Oct-24	0	SK Livestock	Raiwind	No	13-Oct-24	74	0	0	1,115	4,500	1,535	82,510	85,475		82
26		159	13-Oct-24	0	SK Livestock	Raiwind	No	13-Oct-24	75	0	0	1,115	4,500	1,488	83,625	86,637		82
27		355	11-Nov-24	213,000	SK Live Stock	Raiwind	No	11-Nov-24		0	0	600		3,763	0	209,237		111
28		335	27-Nov-24	201,000	local Party	Lodhran	No	27-Nov-24		0	0	600			0	201,000		127
29		358	27-Nov-24	214,800	Local Party	Lodhran	No	27-Nov-24		0	0	600			0	214,800		127
30		365	12-Nov-24		SK Live Stock	Raiwind	No	13-Nov-24	191	0	0	1,100	7,000	3,755	210,100	213,345		112
31		418	24-Nov-24	250,800	Local Party	Lodhran	No	24-Nov-24		0	0	600			0	250,800		124
32		402	12-Nov-24		SK Live Stock	Raiwind	No	13-Nov-24	209	0	0	1,100	7,000	4,136	229,900	232,764		112
33		367	27-Nov-24	220,200	Local Party	Lodhran	No	27-Nov-24		0	0	600			0	220,200		127
34		378	15-Dec-24	220,000	Local Party	Lodhran	No	15-Dec-24		0	0				0	220,000		145
35		382	27-Nov-24	229,200	Local Party	Lodhran	No	27-Nov-24		0	0	600			0	229,200		127
36		341	11-Nov-24	204,600	SK Live Stock	Raiwind	No	11-Nov-24		0	0	600		3,614	0	200,986		111
37		394	15-Dec-24	229,000	Local Party	Lodhran	No	15-Dec-24		0	0				0	229,000		145

Analysis summary | Cleaned Dataset(Insurance data) | Original Dataset(Insurance)

Ready | Accessibility: Investigate | 70%

73°F Clear | Search | 9:18 PM 11/11/2025

Middle section of messy dataset, 37 rows

Data Analysis - Excel

C1 | FARM NAME

	CW	CX	CY	CZ	DA	DB	DC	DD	DE	DF	DG	DH	DI	DJ	DK	DL	DM	DN	DO	DP	DQ	DR	DS
1	Sale	Net Payment	Payment Date	Animal days	Weight gain p	Dressing Rati	Rate Per Kgs			First Sum Insu	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured (	Sum Insured
2		PKR									30-Aug-24	16-Sep-24	30-Sep-24	15-Oct-24	31-Oct-24	15-Nov-24	29-Nov-24	16-Dec-24	31-Dec-24	15-Jan-25	31-Jan-25	17-Feb-25	28-Feb-25	17-M
3	,355	104,193		81	1.1728	48.05%		0		68,680	104,003.27	129,176.86	127,137.92	127,137.92	127,137.92	127,137.92	127,137.92	127,137.92	127,137.92	127,137.92	127,137.92	127,137.92	127,137.92	127,1
4	,000	134,494		96	0.8958	47.62%		0		83,830	108,703.98	134,005.90	125,304.20	134,232.70	134,232.70	134,232.70	134,232.70	134,232.70	134,232.70	134,232.70	134,232.70	134,232.70	134,232.70	134,2
5	,970	106,124		81	1.037	47.91%		0		66,155	105,766.04	130,987.75	117,754.56	117,754.56	117,754.56	117,754.56	117,754.56	117,754.56	117,754.56	117,754.56	117,754.56	117,754.56	117,754.56	117,7
6	,995	128,824		81	0.5432	48.71%		0		94,940	116,342.64	141,853.10	134,472.80	134,472.80	134,472.80	134,472.80	134,472.80	134,472.80	134,472.80	134,472.80	134,472.80	134,472.80	134,472.80	134,4
7	,600	131,797		101	0.8614	51.56%		0		69,690	94,601.85	116,342.64	114,780.96	127,105.30	125,495.34	125,495.34	125,495.34	125,495.34	125,495.34	125,495.34	125,495.34	125,495.34	125,495.34	125,4
8	,800	146,093		98	0.9388	48.48%		0		86,860	115,167.46	140,645.84	136,306.52	144,923.80	144,923.80	144,923.80	144,923.80	144,923.80	144,923.80	144,923.80	144,923.80	144,923.80	144,923.80	144,9
9	,600	132,952		98	0.8673	46.96%		0		81,810	113,992.29	139,438.58	132,027.84	140,766.15	140,766.15	140,766.15	140,766.15	140,766.15	140,766.15	140,766.15	140,766.15	140,766.15	140,766.15	140,7
10	,570	134,174		81	1.0617	46.09%		0		85,850	126,158.71	148,493.03	144,863.88	144,863.88	144,863.88	144,863.88	144,863.88	144,863.88	144,863.88	144,863.88	144,863.88	144,863.88	144,863.88	144,8
11	,100	80,989		95	0.5895	43.83%		0		59,277	77,951.42	92,839.08	77,750.82	91,287.66	91,287.66	91,287.66	91,287.66	91,287.66	91,287.66	91,287.66	91,287.66	91,287.66	91,287.66	91,2
12	0	0		16	-6.375	0.00%		1		56,162	56,162.00	56,162.00	56,162.00	56,162.00	56,162.00	56,162.00	56,162.00	56,162.00	56,162.00	56,162.00	56,162.00	56,162.00	56,162.00	56,1
13	,200	92,900		95	0.8947	45.30%		0		52,317	62,475.77	75,085.56	78,911.28	92,443.20	92,443.20	92,443.20	92,443.20	92,443.20	92,443.20	92,443.20	92,443.20	92,443.20	92,443.20	92,4
14	,740	87,715		82	0.939	46.63%		0		46,313	73,939.21	88,725.95	93,965.76	93,965.76	93,965.76	93,965.76	93,965.76	93,965.76	93,965.76	93,965.76	93,965.76	93,965.76	93,965.76	93,9
15	,200	103,662		95	0.7053	44.88%		0		76,442	95,189.43	108,116.39	105,860.16	119,977.90	119,977.90	119,977.90	119,977.90	119,977.90	119,977.90	119,977.90	119,977.90	119,977.90	119,977.90	119,9
16	,700	76,758		95	0.6737	46.21%		0		43,882	53,646.16	67,061.15	70,788.06	81,670.94	81,670.94	81,670.94	81,670.94	81,670.94	81,670.94	81,670.94	81,670.94	81,670.94	81,670.94	81,6
17	,400	95,229		95	0.7684	50.00%		0		56,040	66,487.98	79,097.76	81,812.43	95,332.05	95,332.05	95,332.05	95,332.05	95,332.05	95,332.05	95,332.05	95,332.05	95,332.05	95,332.05	95,3
18	,660	96,429		82	0.7317	45.41%		0		54,817	83,109.97	98,127.38	107,049.60	107,049.60	107,049.60	107,049.60	107,049.60	107,049.60	107,049.60	107,049.60	107,049.60	107,049.60	107,049.60	107,0
19	,120	100,889		82	0.9756	47.57%		0		65,179	91,076.31	104,003.27	102,886.56	102,886.56	102,886.56	102,886.56	102,886.56	102,886.56	102,886.56	102,886.56	102,886.56	102,886.56	102,886.56	102,8
20	,969	104,924		118	0.8898	47.56%		0		46,169	75,658.73	90,488.72	81,232.20	94,754.28	96,124.56	102,840.51	102,840.51	102,840.51	102,840.51	102,840.51	102,840.51	102,840.51	102,840.51	102,8
21	,200	82,169		95	0.7579	46.75%		0		44,996	64,195.29	76,805.07	78,331.05	91,865.43	91,865.43	91,865.43	91,865.43	91,865.43	91,865.43	91,865.43	91,865.43	91,865.43	91,865.43	91,8
22	,935	79,975		82	0.9634	44.23%		0		43,382	65,341.63	77,951.42	79,491.51	79,491.51	79,491.51	79,491.51	79,491.51	79,491.51	79,491.51	79,491.51	79,491.51	79,491.51	79,491.51	79,4
23	,775	97,590		82	0.9756	47.22%		0		55,202	84,829.49	99,890.15	104,076.00	104,076.00	104,076.00	104,076.00	104,076.00	104,076.00	104,076.00	104,076.00	104,076.00	104,076.00	104,076.00	104,0
24	,280	83,329		82	0.8659	46.45%		0		46,169	72,792.87	85,402.66	83,553.12	83,553.12	83,553.12	83,553.12	83,553.12	83,553.12	83,553.12	83,553.12	83,553.12	83,553.12	83,553.12	83,5
25	,510	85,475		82	0.939	45.12%		0		49,186	63,622.11	76,231.90	90,397.44	90,397.44	90,397.44	90,397.44	90,397.44	90,397.44	90,397.44	90,397.44	90,397.44	90,397.44	90,397.44	90,3
26	,625	86,637		82	0.9268	47.17%		0		45,669	71,073.35	83,683.14	89,802.72	89,802.72	89,802.72	89,802.72	89,802.72	89,802.72	89,802.72	89,802.72	89,802.72	89,802.72	89,802.72	89,8
27	0	209,237		111	1.1441	0.00%		0		137,291	165,912.67	167,110.59	190,587.60	193,388.00	213,657.98	213,657.98	213,657.98	213,657.98	213,657.98	213,657.98	213,657.98	213,657.98	213,657.98	213,6
28	0	201,000		127	0.8661	0.00%		0		140,244	159,923.04	161,120.97	175,461.60	178,512.00	186,073.44	189,156.42	189,156.42	189,156.42	189,156.42	189,156.42	189,156.42	189,156.42	189,156.42	189,1
29	0	214,800		127	0.8976	0.00%		0		146,056	177,292.96	178,490.88	177,276.72	180,297.12	189,606.48	192,627.18	192,627.18	192,627.18	192,627.18	192,627.18	192,627.18	192,627.18	192,627.18	192,6
30	,100	213,345		112	1.1071	52.33%		0		149,978	166,511.63	167,709.56	186,957.36	189,817.76	198,439.08	198,439.08	198,439.08	198,439.08	198,439.08	198,439.08	198,439.08	198,439.08	198,439.08	198,4
31	0	250,800		124	1.1855	0.00%		0		161,977	195,261.84	196,459.77	205,713.60	208,264.00	223,803.75	226,331.10	226,331.10	226,331.10	226,331.10	226,331.10	226,331.10	226,331.10	226,331.10	226,3
32	,900	232,764		112	1.0982	51.99%		0		171,055	188,673.25	189,871.18	221,628.00	225,840.00	233,949.52	233,949.52	233,949.52	233,949.52	233,949.52	233,949.52	233,949.52	233,949.52	233,949.52	233,9
33	0	220,200		127	0.811	0.00%		0		162,352	189,272.21	190,470.14	201,478.32	204,098.72	210,077.12	212,845.05	212,845.05	212,845.05	212,845.05	212,845.05	212,845.05	212,845.05	212,845.05	212,8
34	0	220,000		145	1.1655	0.00%		0		131,479	144,267.62	145,474.88	163,360.80	166,611.20	176,652.00	179,901.06	185,413.86	185,413.86	185,413.86	185,413.86	185,413.86	185,413.86	185,413.86	185,4
35	0	229,200		127	0.6063	0.00%		0		179,476	183,881.55	185,079.47	203,293.44	205,883.84	220,819.70	223,399.35	223,399.35	223,399.35	223,399.35	223,399.35	223,399.35	223,399.35	223,399.35	223,3
36	0	200,986		111	0.991	0.00%		0		142,838	162,917.85	164,115.78	182,722.08	185,652.48	197,850.24	197,850.24	197,850.24	197,850.24	197,850.24	197,850.24	197,850.24	197,850.24	197,850.24	197,8
37	0	229,000		145	1.1379	0.00%		0		138,822	152,136.53	153,334.45	167,596.08	170,776.48	183,129.24	186,264.12	186,594.84	186,594.84	186,594.84	186,594.84	186,594.84	186,594.84	186,594.84	186,5

Analysis summary | Cleaned Dataset(Insurance data) | Original Dataset(Insurance)

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Cleaned Data (After Cleaning)

REPORTING UNIT	FARM NAME	Tag Number	Premium(PKR)	Purchase Date	Purchase month	Tagging Date	Date	Claim Reason (death, theft, accident, disease, etc.)	Reported to the insurance company on (Date)	Farm Exit Date	Live Cattle Sale Amount(PKR)	Buyer Name	Rejection(Yes/NO)
10190	SK Chak Madrassa	1	4,232.45	17-Oct-24	Oct	17-Oct-24				2/4/2025		SK LiveStock	No
10191	SK Chak Madrassa	2	5,011.79	17-Oct-24	Oct	17-Oct-24				1/3/2025		SK LiveStock	No
10192	SK Chak Madrassa	4	3,330.12	17-Oct-24	Oct	17-Oct-24				2/4/2025		SK LiveStock	No
10193	SK Chak Madrassa	6	4,594.18	17-Oct-24	Oct	17-Oct-24				2/18/2025		SK LiveStock	No
10194	SK Chak Madrassa	7	4,196.88	17-Oct-24	Oct	17-Oct-24				1/23/2025	0	SK Livestock	No
10195	SK Chak Madrassa	8	4,054.62	17-Oct-24	Oct	17-Oct-24				2/7/2025		SK LiveStock	No
10197	SK Chak Madrassa	10	4,356.93	17-Oct-24	Oct	17-Oct-24				1/26/2025	149,813	Mandi	No
10198	SK Chak Madrassa	11	3,983.48	17-Oct-24	Oct	17-Oct-24				2/7/2025		SK LiveStock	No
10199	SK Chak Madrassa	12	3,983.48	17-Oct-24	Oct	17-Oct-24				1/26/2025	144,000	Mandi	No
10203	SK Chak Madrassa	16	3,965.70	17-Oct-24	Oct	17-Oct-24				2/7/2025		SK LiveStock	No
10204	SK Chak Madrassa	17	4,196.88	17-Oct-24	Oct	17-Oct-24				1/16/2025	0	SK Livestock	No
10206	SK Chak Madrassa	19	2,311.26	17-Oct-24	Oct	17-Oct-24	16-Nov-24	Leg Fracture					No
10207	SK Chak Madrassa	20	2,128.50	17-Oct-24	Oct	17-Oct-24	24-Oct-24	Chronic Babesiosis and Clostridia Infection					No
10208	SK Chak Madrassa	21	4,614.27	17-Oct-24	Oct	17-Oct-24				3/26/2025		SK LiveStock	No
10209	SK Chak Madrassa	22	4,179.10	17-Oct-24	Oct	17-Oct-24				1/26/2025	156,000	Mandi	No
2545	SK Chak Madrassa	31	3,137.29	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2546	SK Chak Madrassa	32	2,265.53	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2547	SK Chak Madrassa	33	2,500.47	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2548	SK Chak Madrassa	34	3,258.62	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2549	SK Chak Madrassa	35	2,399.78	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2550	SK Chak Madrassa	36	2,164.84	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2551	SK Chak Madrassa	37	2,617.30	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2552	SK Chak Madrassa	38	2,842.63	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2553	SK Chak Madrassa	39	2,981.29	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2544	SK Chak Madrassa	40	3,449.29	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	Yes
2554	SK Chak Madrassa	41	2,998.63	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2555	SK Chak Madrassa	42	3,102.62	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2556	SK Chak Madrassa	43	3,720.69	14-Aug-24	Aug	14-Aug-24				10/13/2024	0	SK LiveStock	No
2557	SK Chak Madrassa	44	3,171.96	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2558	SK Chak Madrassa	45	2,998.63	14-Aug-24	Aug	14-Aug-24				10/26/2024		SK LiveStock	No
2559	SK Chak Madrassa	46	4,160.57	14-Aug-24	Aug	14-Aug-24				10/13/2024	0	SK LiveStock	No
2560	SK Chak Madrassa	47	4,839.71	14-Aug-24	Aug	14-Aug-24				10/13/2024	0	SK LiveStock	No
2925	SK Chak Madrassa	48	3,628.50	16-Aug-24	Aug	16-Aug-24				12/21/2024	0	SK LiveStock	No
2926	SK Chak Madrassa	49	3,965.70	16-Aug-24	Aug	16-Aug-24				2/4/2025		SK LiveStock	No
2927	SK Chak Madrassa	50	3,681.60	16-Aug-24	Aug	16-Aug-24				12/21/2024	0	SK LiveStock	No

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Data Analysis - Excel

J1 | Farm Exit Date

	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Claim Reason (death, theft, accident, disease, etc.)	Farm Exit Date	Live Cattle Sale Amount(PKR)	Buyer Name	Rejection(Yes/NO)	Sale Month	Sale Date	Carcass Sale Amount(PKR)	Net Payment(	Payment Date	Animal days	Rate Per Kgs	First Sum Insured	Sum Insured Dated(30-Apr-25)
2		2/4/2025		SK LiveStock	No	Feb	2/6/2025	149,964	152,006		111		92,400	141,081.64
3		1/3/2025		SK LiveStock	No	Jan	1/6/2025	155,496	157,086		79		103,600	167,059.62
4		2/4/2025		SK LiveStock	No	Feb	2/6/2025	141,178	143,375		111		90,720	111,003.95
5		2/18/2025		SK LiveStock	No	Feb	2/18/2025	143,190	145,068		125		77,550	153,139.20
6		1/23/2025	0	SK Livestock	No	Jan	1/23/2025	138,259	140,043		99		86,900	139,896.08
7		2/7/2025		SK LiveStock	No	Feb	2/6/2025	136,329	138,417		114		74,250	135,153.84
8		1/26/2025	149,813	Mandi	No	Jan	1/26/2025	0	149,813		102		84,150	145,231.10
9		2/7/2025		SK LiveStock	No	Feb	2/6/2025	137,490	139,573		114		71,500	132,782.72
10		1/26/2025	144,000	Mandi	No	Jan	1/26/2025	0	144,000		102		75,350	132,782.72
11		2/7/2025		SK LiveStock	No	Feb	2/6/2025	140,223	142,266		114		70,400	132,189.94
12		1/16/2025	0	SK Livestock	No	Jan	1/23/2025	147,766	149,557		92		96,880	139,896.08
13	Leg Fracture				No	Jan	1/1/2025	0	0		31		69,300	77,042.00
14	Chronic Babesiosis and Clostridia Infection				No	Jan	1/1/2025	0	0		8		70,950	70,950.00
15		3/26/2025		SK LiveStock	No	Mar	3/26/2025	162,735	158,736		161		67,100	153,809.00
16		1/26/2025	156,000	Mandi	No	Jan	1/26/2025	0	156,000		102		71,500	139,303.30
17		10/26/2024		SK LiveStock	No	Oct	10/27/2024	93,500	96,270		74		70,490	104,576.37
18		10/26/2024		SK LiveStock	No	Oct	10/27/2024	72,600	75,678		74		50,350	75,517.65
19		10/26/2024		SK LiveStock	No	Oct	10/27/2024	78,100	81,029		74		56,710	83,349.11
20		10/26/2024		SK LiveStock	No	Oct	10/27/2024	101,200	103,821		74		62,540	108,620.76
21		10/26/2024		SK LiveStock	No	Oct	10/27/2024	83,600	86,569		74		53,530	79,992.77
22		10/26/2024		SK LiveStock	No	Oct	10/27/2024	80,300	83,517		74		55,120	72,161.31
23		10/26/2024		SK LiveStock	No	Oct	10/27/2024	80,300	83,289		74		63,600	87,243.27
24		10/26/2024		SK LiveStock	No	Oct	10/27/2024	90,200	93,039		74		51,940	94,754.28
25		10/26/2024		SK LiveStock	No	Oct	10/27/2024	94,600	97,241		74		58,300	99,376.44
26		10/26/2024		SK LiveStock	Yes	Oct	10/27/2024	110,100	112,432		74		71,020	114,976.23
27		10/26/2024		SK LiveStock	No	Oct	10/27/2024	82,500	85,359		74		68,370	99,954.21
28		10/26/2024		SK LiveStock	No	Oct	10/27/2024	96,800	99,411		74		66,250	103,420.83
29		10/13/2024	0	SK LiveStock	No	Oct	10/13/2024	118,190	121,028		61		83,160	124,022.85
30		10/26/2024		SK LiveStock	No	Oct	10/27/2024	108,900	111,421		74		76,850	105,731.91
31		10/26/2024		SK LiveStock	No	Oct	10/27/2024	95,700	98,410		74		68,370	99,954.21
32		10/13/2024	0	SK LiveStock	No	Oct	10/13/2024	134,915	138,953		61		95,580	138,685.65
33		10/13/2024	0	SK LiveStock	No	Oct	10/13/2024	144,950	148,548		61		125,820	161,323.68
34		12/21/2024	0	SK LiveStock	No	Dec	12/24/2024	125,400	128,221		128		45,580	120,950.00
35		2/4/2025		SK LiveStock	No	Feb	2/6/2025	136,688	139,205		173		49,290	132,189.94
36		12/21/2024	0	SK LiveStock	No	Dec	12/24/2024	123,200	126,021		128		47,700	122,720.00

Analysis summary | Cleaned Dataset(Insurance data) | Original Dataset(Insurance)

Right half of cleaned dataset, 37 rows

Analysis Summary & Result

Data Analysis - Excel

Total premium collected Per Each Farm

Farm name	HI Cattles Farm	Baloch Cattle Farn	Organico Farm	Quality Meat E	Rohi Cattle	Sindhu Cattle Farn	SK - Duniyapur	SK Chak Madrassa	SK Live Stock Balloki	SK Oasis Feedlot farm	Sum total premium
Total premium collected	8,110,757.64	5,821,481.65	10,395,394.12	2,765,715.93	3,759,194.85	6,742,165.09	15,297,525.48	7,756,126.90	7,671,282.34	15,181,986.80	83,501,630.81

Yearly Analysis

Net payment year	2024	2025	Total
Total net payment per year(PKR)	817,698,450	2,165,358,152	2,983,056,602

2025 has higher business volume

2025 claims exceed 2024 by 1,347,659,702

Highest Initial Sum Insured

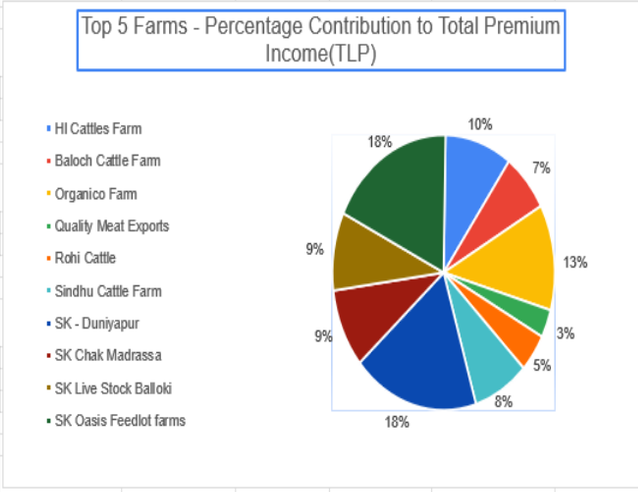
Farm name	Baloch Cattle Farm
Sum Insured	189,488

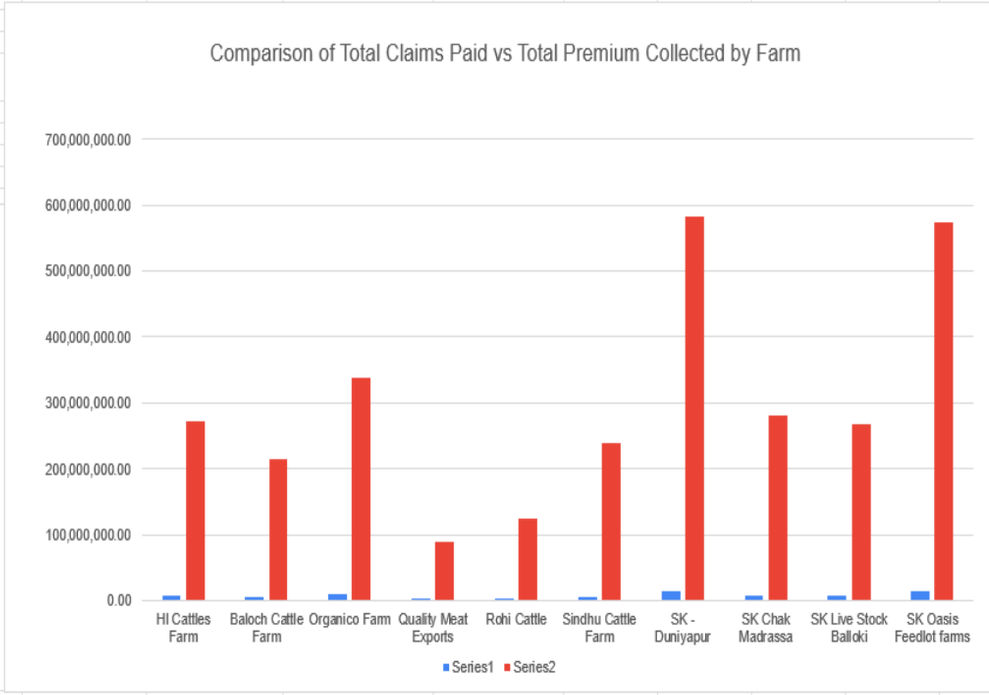
Highest tag number

Farm name	HI Cattles Farm
Tag Number	LSS-133348

Highest premium collected

Farm name	SK Oasis Feedlot farms
Tag Number	LSS-16218
Premium collected	9,046.29





Total Net payment per Each farm

Farm name	HI Cattles Farm	Baloch Cattle Farn	Organico Farm	Quality Meat E	Rohi Cattle	Sindhu Cattle Farn	SK - Duniyapur	SK Chak Madrassa	SK Live Stock Balloki	SK Oasis Feedlot farm	Sum total net payment
Net payment	273,077,039	213,993,204	338,966,296	89,952,832	124,222,502	239,282,982	583,223,578	280,029,855	266,928,930	573,379,384	2,983,056,602

Best claim to premium ratio

Farm name	HI Cattles Farm	Baloch Cattle Farn	Organico Farm	Quality Meat E	Rohi Cattle	Sindhu Cattle Farn	SK - Duniyapur	SK Chak Madrassa	SK Live Stock Balloki	SK Oasis Feedlot farms

Analysis summary | Cleaned Dataset(Insurance data) | Original Dataset(Insurance)

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Project Summary Insurance Claims Data Analysis

This project involved cleaning, transforming, and analyzing a large-scale insurance dataset containing over 22,000 records and 104 columns collected from multiple farm branches to assess portfolio performance and risk exposure. The analysis focused on evaluating premium trends, claims behavior, and farm-level efficiency using claims-to-premium ratios. Through systematic data cleaning, missing-value treatment, and formula-based computations, farm-level total premiums, yearly net payments, and claims-to-premium ratios were calculated.

The results showed that SK Oasis Feedlot Farms generated the highest total premium collection, while Baloch Cattle Farm recorded the highest initial sum insured. Year-over-year comparisons indicated higher overall business volume in 2025 compared to 2024. However, the claims-to-premium comparison revealed meaningful variation in farm-level efficiency, indicating that higher premium contribution does not necessarily imply stronger profitability across all farms.

To support decision-making, a 2D pie chart was used to highlight revenue concentration among the top five farms, and a bar chart compared total claims paid against total premium collected. These visualizations enable management to quickly assess revenue dependency and evaluate whether claims behavior aligns with premium contributions across the portfolio.

Management Recommendations

Based on the claims-to-premium analysis and year-over-year comparison, the portfolio shows clear differences in farm-level risk and revenue contribution. Farms with relatively high claims compared to premium income should be prioritized for underwriting review and potential premium reassessment to protect overall profitability. The concentration of premium income among a small number of top farms highlights exposure to revenue dependency risk, indicating a need for gradual portfolio diversification. Although business volume increased in 2025 compared to 2024, premium growth should continue to be evaluated alongside claims behavior to ensure sustainable and profitable expansion. Regular monitoring of claims-to-premium ratios would support early risk identification and informed pricing and portfolio management decisions.